WORKED SOLUTION

STEP III 2004, Question 3

Pure • Convexity and Series Bounds

STEP III 2004, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q3

The key idea

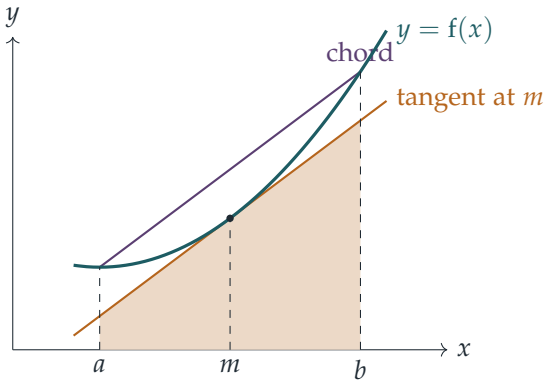
If $\mathrm{f}'' > 0$ the curve bends upwards. So it lies above every tangent line and below every chord. The tangent at the midpoint and the chord from end to end each cut off a trapezium whose area is easy to write down, and these two areas trap the integral. With $\mathrm{f}(x) = 1/x^2$ on $[n-1, n]$ the integral telescopes. Adding the inequalities over n then gives bounds on $\sum 1/n^2$.

Solution

The two bounds on the integral

Write $m = \frac{1}{2}(a+b)$. Since $\mathrm{f}'' > 0$ on $[a,b]$, the gradient f' is strictly increasing, so the curve is convex. This gives two facts.

- *The curve lies above its tangent at m.* Let $t(x) = \mathrm{f}(m) + \mathrm{f}'(m)(x-m)$ and $g = \mathrm{f} - t$. Then $g(m) = g'(m) = 0$ and $g'' = \mathrm{f}'' > 0$, so g' is increasing. Hence $g' < 0$ to the left of m and $g' > 0$ to the right, so g has a strict minimum 0 at m, and $\mathrm{f}(x) > t(x)$ for $x \neq m$.
- *The curve lies below its chord.* Let $c(x)$ be the straight line through $(a, \mathrm{f}(a))$ and $(b, \mathrm{f}(b))$, and $h = \mathrm{f} - c$. Then $h(a) = h(b) = 0$ and $h'' > 0$. If h had a value ≥ 0 inside (a,b), it would have a maximum there with $h'' \leq 0$, which is impossible. So $\mathrm{f}(x) < c(x)$ for $a < x < b$.



Now compare areas over $[a,b]$. The area under the tangent is a trapezium of width $b-a$. Its mean height is the height of t at the midpoint, $t(m)=\mathrm{f}(m)$, so the area is $(b-a)\,\mathrm{f}(m)$ (shaded). The area under the chord is a trapezium with parallel sides $\mathrm{f}(a)$ and $\mathrm{f}(b)$, so the area is $(b-a)\frac{1}{2}\big(\mathrm{f}(a)+\mathrm{f}(b)\big)$. The curve lies strictly between the two lines except at isolated points. So

$$(b-a)\,\mathrm{f}\Big(\frac{a+b}{2}\Big) < \int_a^b \mathrm{f}(x)\,\mathrm{d}x < (b-a)\,\frac{\mathrm{f}(a)+\mathrm{f}(b)}{2}.$$

(The sketch shows a positive f. The argument is really about $\int(\mathrm{f}-t)>0$ and $\int(c-\mathrm{f})>0$, so the sign of f doesn't matter.)

Choosing a, b and f

The middle term $\frac{1}{n-1}-\frac{1}{n}$ looks like $\big[-\frac{1}{x}\big]_{n-1}^{n}$, so take $\mathrm{f}(x)=1/x^2$, $a=n-1$, $b=n$. Then $b-a=1$, and $\mathrm{f}''(x)=6/x^4>0$ on $[n-1,n]$ because $n-1\geq 1>0$. Also

$$\int_{n-1}^{n}\frac{\mathrm{d}x}{x^2}=\frac{1}{n-1}-\frac{1}{n}, \qquad \mathrm{f}\Big(\frac{2n-1}{2}\Big)=\frac{4}{(2n-1)^2},$$

so the first part gives

$$\frac{4}{(2n-1)^2} < \frac{1}{n-1}-\frac{1}{n} < \frac{1}{2}\Big(\frac{1}{n^2}+\frac{1}{(n-1)^2}\Big) \qquad (n\geq 2),$$

as required.

Deducing the two bounds on 1

Write $S=\sum_{n=1}^{\infty}\frac{1}{n^2}$. This series converges, because for $n\geq 2$ we have $\frac{1}{n^2}<\frac{1}{n(n-1)}=\frac{1}{n-1}-\frac{1}{n}$, and those terms telescope. Add the inequalities for $n=2,3,\ldots,N$. The middle terms telescope to $1-\frac{1}{N}$, which tends to 1.

Ask yourself When we let $N \to \infty$, strict inequalities usually become $\leq$. How do we keep them strict?

Look at the gaps. For each $n \geq 2$ the gap $d_n = \left(\frac{1}{n-1} - \frac{1}{n}\right) - \frac{4}{(2n-1)^2}$ is positive. So the infinite sum of the gaps is at least its first term, $d_2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} > 0$. Every series here converges, so we may sum term by term, and the difference of the sums is strictly positive. The same argument works for the right-hand inequality.

On the left, $n = 2, 3, 4, \ldots$ gives $2n - 1 = 3, 5, 7, \ldots$, so

$$4\left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots\right) < 1.$$

On the right,

$$\sum_{n=2}^{\infty} \frac{1}{2}\left(\frac{1}{n^2} + \frac{1}{(n-1)^2}\right) = \frac{1}{2}(S-1) + \frac{1}{2}S = \frac{1}{2} + (S-1),$$

which gives

$$1 < \frac{1}{2} + \left(\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots\right).$$

Comparing with the odd squares

Group the left-hand series in consecutive pairs, an odd term followed by an even term. For $k \geq 2$,

$$\frac{1}{2}\left(\frac{1}{(2k-1)^2} + \frac{1}{(2k)^2}\right) < \frac{1}{2}\left(\frac{1}{(2k-1)^2} + \frac{1}{(2k-1)^2}\right) = \frac{1}{(2k-1)^2},$$

since $2k > 2k - 1 > 0$. With $k = 2, 3, 4, \ldots$ the pairs on the left are $(3^2, 4^2)$, $(5^2, 6^2), \ldots$, which together use every term $\frac{1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \ldots$ exactly once. (Grouping is allowed because all the terms are positive and the series converges.) The right-hand sides are $\frac{1}{3^2}, \frac{1}{5^2}, \ldots$. Summing over k, with strictness kept as before, gives

$$\frac{1}{2}\left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \cdots\right) < \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots.$$

The bounds on S

Lower bound. From the right-hand inequality, $1 < \frac{1}{2} + (S-1)$, so $S > \frac{3}{2}$.

Upper bound. Let $T = \frac{1}{3^2} + \frac{1}{4^2} + \cdots = S - 1 - \frac{1}{4}$. Combining the last comparison with the left-hand inequality gives

$$\frac{1}{2}T < \frac{1}{3^2} + \frac{1}{5^2} + \cdots < \frac{1}{4},$$

so $T < \frac{1}{2}$. Hence $S = 1 + \frac{1}{4} + T < \frac{7}{4}$. Altogether

$$\frac{3}{2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \frac{7}{4}.$$

Checks

- The true value is $S = \pi^2/6 \approx 1.6449$, which lies inside $(1.5, 1.75)$.
- Numerically, $4\sum_{k\geq 2}(2k-1)^{-2} = 4(\pi^2/8 - 1) \approx 0.935 < 1$, and $\frac{1}{2} + (S-1) \approx 1.145 > 1$.
- For the comparison, $\frac{1}{2}T \approx 0.197 < 0.234 \approx \sum_{k\geq 2}(2k-1)^{-2}$.
- Try $n = 2$: $\frac{4}{9} \approx 0.444 < \frac{1}{2} < \frac{5}{8} = 0.625$. A check over $2 \leq n \leq 2000$ with exact fractions confirms all three inequalities.

Common slips

- Forgetting the missing $\frac{1}{4}$: the bracket starts at $\frac{1}{3^2}$, so $T = S - \frac{5}{4}$, not $S - 1$. Using $S - 1$ leads to the false claim $S < \frac{3}{2}$.
- Getting the inequalities the wrong way round. For a curve that bends upwards, the tangent (midpoint) estimate is too small and the chord (trapezium) estimate is too big.
- Passing to infinite sums and quietly keeping "$<$" without saying why it survives the limit.

Beyond STEP

The two bounds are the midpoint rule and the trapezium rule. Convexity tells you which way each one errs, and Taylor's theorem shows that both errors shrink like $(b-a)^3$. The exact value $\sum 1/n^2 = \pi^2/6$ is Euler's solution of the Basel problem. A first analysis course proves it with Fourier series. It also makes the "strict inequalities in the limit" step rigorous. Abbott's *Understanding Analysis* covers both convergence of series and convex functions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.